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Estimation Based Control Strategies for an Alcoholic Fermentation Process with Unknown Inputs

Emil Petre, *Member*, *IEEE*
Department of Automatic Control and Electronics
University of Craiova
Craiova, Romania
epetre@automation.ucv.ro

Dorin Şendrescu, *Member*, *IEEE*
Department of Automatic Control and Electronics
University of Craiova
Craiova, Romania
dansel@automation.ucv.ro

Dan Selişteanu, *Member*, *IEEE*
Department of Automatic Control and Electronics
University of Craiova
Craiova, Romania
dansel@automation.ucv.ro

Monica Roman, *Member*, *IEEE*
Department of Automatic Control and Electronics
University of Craiova
Craiova, Romania
monica@automation.ucv.ro

***Abstract*—The present paper focuses on estimation and advanced control structures for an alcoholic fermentation process (AFP) exploited for the ethanol production. Taking into account the inhibition problems, the control aim consists in the preservation of the so-called process inhibitory compounds at a prescribed reduced level in spite of large variations of the influent pollutant concentration and of the load, having as a result a large production of alcohol. Using a strongly nonlinear and uncertain model and realistic operating conditions, novel estimation and advanced control schemes are proposed. More precisely, adaptive control laws designed via novel nonlinear estimation algorithms for unknown inputs and kinetics are developed. To illustrate that the control goal is fulfilled, various tests performed by computer simulations under realistic conditions are included.**

Keywords—nonlinear systems, observers, adaptive control, alcoholic fermentation processes

I. INTRODUCTION

The growth of world human population as well as of its prosperity has led to a sharp increasing of the world energy consumption. Taking into account that the reserves of traditional non-renewable energy sources (fossil fuels) are limited, in recent years, the use of some renewable sources (wind, solar energy, energy from biomass, geothermal energy, hydropower, etc.) for energy production has gained a great attention [1]-[3].

Besides the technology development for biogas production [1], [2], [4], [5], a particular attention was paid to the biomass conversion (which is seen as one of the most renewable energy resources) into fuel ethanol [3], [6]-[8]. Usually, in industry, these processes occur in Continuous Stirred Tank Reactors (CSTRs) and can be modelled by ordinary differential equations (ODEs), which express the mass and energy balances. One drawback associated with the continuous fermentation is the unusual occurrence of oscillation under certain fermentation conditions, i.e., these can occur in periods of time during which the ethanol productivity is decreased, and then a high level of residual substrate is loosed [8], [9].

To improve the ethanol production by fermentation, the researches were oriented toward the modelling of this complex bioprocess and to find competitive ethanologenic bacteria (biomass) able to convert several kinds of substrates [3], [7], [8]. Also, very important are the monitoring and control strategies [3], [8], able to assure an effective implementation and operation of such complex processes, an essential engineering challenge being the enhancement of production and the bioprocess stability. Nevertheless, the modern design of useful control techniques for this kind of processes is obstructed by the existence of uncertain and time varying yield coefficients and kinetic parameters of the process models [3], [6]-[8]. Another key problem is to find proper and inexpensive instrumentation for the measurement of the process biological variables used in the control schemes [3], [6]-[10]. To cope with these problems, various control methods were designed: nonlinear controllers [11], nonlinear predictive techniques [12], [13], optimal control [14], several numerical methods [6], [7], robust-adaptive strategies [15], etc. Several techniques require the use of "software sensors" [10] used for the estimation of unavailable states implied in control [3], [6]-[8] and furthermore for the estimation of kinetic reactions [4], [5].

This work is dedicated to the development of two innovative adaptive control schemes for a complex AFP taking place in a CSTR. The proposed control structure is able to ensure a large production of alcohol together with a minimal quantity of residual substrate. Moreover, the process instability (that can appear because of unsuitable process operation or due to disturbances caused by the load rate variations) is avoided [6]-[8]. It must be mentioned that several aspects about the AFP control were presented in [15]. However, in [15] the control design was realised supposing that the evolution of the influent substrate concentration is known.

In the present work, the control design is done assuming that the influent substrate concentration is unknown (as in the real plants) and the bioprocess kinetics is time varying and unknown. The adaptive structures developed in this work include a linearizing control law (LCL) in which all the unknown variables (state variables, kinetics) are replaced by their on-line estimations obtained via suitable state observers and parameter estimators. A sliding mode observer (SMO) reconstructs the unavailable input concentration from the alcohol and substrate concentrations measured inside the reactor. The parameter estimator provides the unknown process kinetics using the on-line information supplied by the previous observer.

The present paper is structured in the following manner. After the presentation of the alcoholic fermentation process and of its model in Section II, the next section approaches the proposed nonlinear estimation and adaptive control strategies. Section IV is dedicated to realistic simulations which allow the analysis of the closed-loop process behaviour. Some conclusions are delivered in Section V.

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II. Fermentation Process Description and Modelling

The bioprocess which will be analysed is a continuous alcoholic fermentation of *Zymomonas Mobilis* bacteria, which is used to produce ethanol [6]-[8], [15]. It was considered that these bacteria possess superior characteristics compared to those of yeast for industrial production of ethanol [8]. The model is described by the following equations [6]-[8]:

$$\dot{C}_X(t) = \mu_X(\cdot)C_X + (rd/4 + d_S)\cdot(r-1)C_X, \quad (1)$$

$$\dot{C}_S(t) = (-1/Y_{PS})\cdot\nu_P(\cdot)C_X + d_S C_{Sin} - d_S C_S, \quad (2)$$

$$\dot{C}_P(t) = \nu_P(\cdot)C_X - d_S C_P, \quad (3)$$

$$\dot{Z}(t) = \gamma\cdot(I - Z), \quad (4)$$

$$\dot{I}(t) = \gamma\cdot(\nu_P(\cdot)C_X - d_S C_P - I) \quad (5)$$

where C_X, C_S, C_{Sin}, C_P are, respectively, the biomass, the substrate, the influent substrate and the product (ethanol) concentrations, I is an auxiliary variable used to express the inhibition effect, and Z is the average of ethanol concentration change rate. The biomass specific growth rate is denoted with μ_X, r is the microorganisms recycling rate, and $d = d_S + d_r$ is the so-called total dilution rate (where d_S and d_r are the substrate and, respectively, recycled biomass dilution rates). Y_{PS} is called the yield coefficient of product on substrate, ν_P is ethanol production specific rate, and γ is the weighted historic parameter for C_P change rate.

The dynamical effect of Z on the cell growth is expressed as [6]-[8]:

$$f_\mu(Z(t)) = (1/2)\cdot\left(1 - \left(e^{\alpha Z-\beta} - e^{-\alpha Z-\beta}\right)/\left(e^{\alpha Z-\beta} + e^{-\alpha Z-\beta}\right)\right), \quad (6)$$

where the parameters α and β are related with the intensity of the inhibition factor of the $Z(t)$ concentration rate [6]-[8].

The instantaneous cell specific growth rate can be expressed by the Veeramallu & Agrawal relations [6]-[8]:

$$\mu(C_S, C_P) = \frac{\mu_{\max} C_S\left(1 - \left(\frac{C_P}{P_{ma}}\right)^a\right)\cdot\left(1 - \left(\frac{C_P - P_{ob}}{P_{mb} - P_{ob}}\right)^b\right)}{K_S + C_S + C_S(C_S - S_i)/(K_i - S_i)}, \quad (7)$$

In (7) $\mu_{\max}$ is the maximum specific growth rate, P_{mb} represents the maximum product inhibition factor for cell growth, P_{ma} and P_{ob} are ethanol inhibition thresholds for $\mu(C_S, C_P)$ [8], a and b are inhibition parameters for C_P production rate, K_S and K_i are the so-called saturation and inhibition coefficients, and S_i is the lower limit of substrate inhibition. In (7) the next constraints take place [8]:

$$\begin{aligned} &(C_P - P_{ob})/(P_{mb} - P_{ob}) = 0, \quad if\ C_P \le P_{ob};\\ &(C_P - P_{ob})/(P_{mb} - P_{ob}) = 1, \quad if\ C_P > P_{mb}\\ &(C_S - S_i) = 0, \qquad if\ C_S \le S_i \end{aligned} \quad (8)$$

The cell specific growth rate has a dynamic defined by:

$$\mu_X(C_S, C_P, Z) = f_\mu(Z(t))\cdot\mu(C_S, C_P). \quad (9)$$

where $f_\mu(Z(t)) \in (0, 1)$ is considered as inhibition factor.

The ethanol specific production rate is given by [6]-[8]:

$$\nu_P(C_S, C_P) = \nu_{P\max}\left(C_S/(K_{mp} + C_S)\right)\cdot\left(1 - (C_P/P_{me})^\delta\right), \quad (10)$$

with $\nu_{P\max}$, K_{mp}, P_{me} and δ, respectively, the maximum specific rate, the saturation coefficient of the substrate, the maximum alcohol inhibition factor and the ethanol inhibition exponent for ethanol production rate.

The expressions (9) and (10) include the effects of the inhibition produced by the substrate and alcohol on the growth and on the production rates. This fact is an indicator of the interaction between growth and production [6]-[8]. The AFP described in equations (1)-(10) can assure a high production of ethanol, but in the same time it can present oscillations of the biological state variables [8], [9]. Considering the inhibition aspects (especially at high substrate concentration), the consequence is that the alcohol production process needs to preserve the substrate concentration C_S in the bioreactor at a required reference C_{Sref}, regardless of influent concentration and load variations. The substrate dilution rate d_S is the control input. As a result, a low level of residual substrate ensures both a large production of alcohol and an adequate yield.

III. Control Design

A. Exact Linearizing Control Law (LCL)

Initially, we will present an ideal case: the process is completely known. This will be considered the benchmark case, i.e. the specific rates μ and ν_P are known, all the states and all the inflow rates are on-line available. Due to the fact that the equation (2) in the process model has the relative order equal to one, then, the next exact LCL

$$d_S = 1/(C_{Sin} - C_S)\cdot\left(\eta\cdot(C_{Sref} - C_S) + (1/Y_{PS})\nu_p C_X\right), \quad (11)$$

assures the stability of the closed-loop system described by the following first order linear stable ODE:

$$(\dot{C}_{Sref} - \dot{C}_S) + \eta\cdot(C_{Sref} - C_S) = 0,\ \eta > 0. \quad (12)$$

where η is a design parameter.

The controller (11) leads to $\dot{\tilde{C}}_S = -\eta\cdot\tilde{C}_S$, where $\tilde{C}_S = C_{Sref} - C_S$. This linear error model is characterized by an exponential stable point at $\tilde{C}_S = 0$ (for $\eta > 0$). The LCL can be utilized as backbone to design some adaptive controllers.

B. New Adaptive Control Strategies

Because the full prior knowledge which was supposed before is not realistic, we will consider more harsh conditions about the bioprocess kinetics, the state variables and the process inputs, as following:

- the influent substrate concentration C_{Sin} is unknown;
- the specific growth rates ν_P and μ_X are completely unknown (and time-varying);
- the biological state variables C_X, I and Z are not accessible;
- the only available on-line measurements are C_S and C_P;
- the rest of process parameters are assumed to be known.

Under these severe conditions, an innovative adaptive control scheme will be developed. To this end, the exact LCL (11) will be used. This becomes an adaptive control law if the unknown and inaccessible variables are substituted by their on-line estimates given by appropriate state observers and parameter estimators.

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Estimation of S_{in}. To estimate the influent concentration S_{in} based on the measurements of C_S and C_P, we propose a particular sliding mode observer (SMO) [16]-[19] achieved from the *generalized super-twisting observer* developed in [20]. To build this estimator, a linear combination of the two available states will be considered as: $\psi = C_S + (1/Y_{PS})\cdot C_P$. From (2) and (3), the dynamic of ψ can be expressed by the linear differential equation:

$$\dot{\psi} = -d_S\,\psi + d_S\,C_{Sin}\,. \tag{13}$$

Based on (13), the SMO able to compute the on-line estimates $\hat{C}_{Sin}$ is described by:

$$\begin{aligned}\dot{\hat{\psi}} &= -d_S\,\psi + d_S\,\hat{C}_{Sin} - \lambda_\psi \kappa_1 \phi_{\psi 1}(\tilde{\psi}),\\ \dot{\hat{C}}_{Sin} &= -\lambda_\psi^2 \kappa_2\, d_S\, \phi_{\psi 2}(\tilde{\psi}), \quad \lambda_\psi,\, \kappa_1,\, \kappa_2 > 0,\end{aligned} \tag{14}$$

where $\tilde{\psi} = \hat{\psi} - \psi$, and:

$$\begin{aligned}\phi_{\psi 1}(\tilde{\psi}) &= \gamma_1 \,|\,\tilde{\psi}\,|^{1/2}\,\mathrm{sign}(\tilde{\psi}) + \gamma_2\tilde{\psi},\\ \phi_{\psi 2}(\tilde{\psi}) &= \frac{\gamma_2^2}{2}\mathrm{sign}(\tilde{\psi}) + \frac{3}{2}\gamma_1\gamma_2\,|\,\tilde{\psi}\,|^{1/2}\,\mathrm{sign}(\tilde{\psi}) + \gamma_2^2\tilde{\psi},\end{aligned} \tag{15}$$

with $\gamma_1, \gamma_2 > 0$. The observer (14), (15) is independent of the process kinetics and it has 5 tuning parameters $\lambda_\psi, \kappa_1, \kappa_2, \gamma_1, \gamma_2$, whose values, usually, are chosen via a trial and error procedure.

Estimation of specific reaction rates. As it was mentioned above, the two bioprocess specific growth rates ν_P and μ_X are time varying and completely unknown. However, the LCL (11) includes only the unknown specific growth rate ν_P. As a result, we will develop some estimators in order to reconstruct the unknown information regarding ν_P. Taking into account that C_X is not available, the substrate consumption rate $-\nu_P(t)\cdot C_X$ from (11) can be expressed as in the next cases.

Case 1. $-\nu_P(t)\cdot C_X = \theta_1(t)\cdot C_S$, where

$$\theta_1(t) = \nu_{P\max}\left(C_X/(K_{mp} + C_S)\right)\cdot\left(1 - (C_P/P_{me})^{\delta}\right)$$

is regarded as an unknown and time-varying parameter. Then, the exact LCL (11) turns into:

$$d_S = 1/(C_{Sin} - C_S)\cdot\left(\eta\cdot(C_{Sref} - C_S) + (1/Y_{PS})\cdot C_S\cdot\theta_1(t)\right). \tag{16}$$

Case 2. The whole reaction rate is presumed to be an unknown parameter $-\nu_P(t)\cdot C_X = \theta_2(t)$, and now the controller (11) becomes:

$$d_S = 1/(C_{Sin} - C_S)\cdot\left(\eta\cdot(C_{Sref} - C_S) + (1/Y_{PS})\cdot\theta_2(t)\right). \tag{17}$$

The unknown parameters $\theta_1(t)$ and $\theta_2(t)$ will be substituted by the on-line estimates $\hat{\theta}_1(t)$ and $\hat{\theta}_2(t)$ obtained from two parameter estimators developed using (2), rewritten as:

$$\dot{C}_S(t) = (-1/Y_{PS})\cdot C_S\cdot\theta_1(t) + d_S C_{Sin} - d_S C_S\,, \tag{18}$$

respectively,

$$\dot{C}_S(t) = (-1/Y_{PS})\cdot\theta_2(t) + d_S C_{Sin} - d_S C_S\,. \tag{19}$$

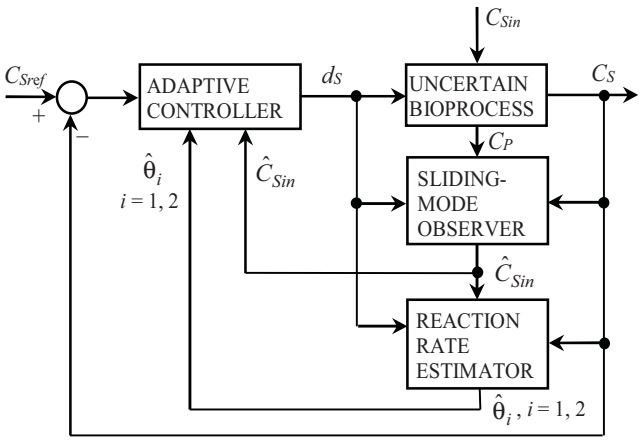


Fig. 1. Diagram of the adaptive closed-loop system.

The proposed estimation algorithm for $\hat{\theta}_1(t)$ is an observer-based estimator (OBE) [10], obtained from (18):

$$\hat{C}_S(t) = \frac{1}{Y_{PS}}C_S\cdot\hat{\theta}_1(t) + d_S\hat{C}_{Sin} - d_S C_S + \lambda_1(C_S - \hat{C}_S)\,, \tag{20}$$

$$\dot{\hat{\theta}}_1(t) = (-1/Y_{PS})\cdot C_S\cdot\omega_1\cdot(C_S - \hat{C}_S)\,. \tag{21}$$

Similarly, by using (19), the on-line estimation algorithm for $\hat{\theta}_2(t)$ is also an OBE, described by:

$$\hat{C}_S(t) = (-1/Y_{PS})\cdot\hat{\theta}_2(t) + d_S\hat{C}_{Sin} - d_S C_S + \lambda_2(C_S - \hat{C}_S)\,, \tag{22}$$

$$\dot{\hat{\theta}}_2(t) = (-1/Y_{PS})\cdot\omega_2\cdot(C_S - \hat{C}_S)\,, \tag{23}$$

In (20)-(23), the estimation $\hat{C}_{Sin}$ is provided by the SMO (14), and the estimators design parameters λ_1, λ_2 and ω_1, ω_2 are chosen so that to assure their stability and tracking properties. So, the stability is assured if these parameters are chosen as, $\lambda_1, \lambda_2 < 0$ and $\omega_1, \omega_2 > 0$ (see [4], [10]). Usually, their values are chosen via trial and error method.

Taking into consideration these two cases, we obtained two adaptive control structures. The first variant of the adaptive controller is derived from the combination of the SMO (14) with the reaction rate observer (20) and (21), and with the controller (16) written as:

$$d_S = 1/(\hat{C}_{Sin} - C_S)\cdot\left(\eta\cdot(C_{Sref} - C_S) + (1/Y_{PS})\cdot C_S\cdot\hat{\theta}_1(t)\right), \tag{24}$$

while the second variant is obtained as a mixture of the SMO (14) with the reaction rate observer (22) and (23) and with the controller (17) rewritten as:

$$d_S = 1/(\hat{C}_{Sin} - C_S)\cdot\left(\eta\cdot(C_{Sref} - C_S) + (1/Y_{PS})\cdot\hat{\theta}(t)\right). \tag{25}$$

Remark 1. For the estimation of the parameters $\theta_1(t)$ and $\theta_2(t)$ the dynamic of variable C_S (and not of variable C_P) is used. Thus, the persistency signal C_{Sin} is included, even if the two parameter estimators used its estimated values.

The block schemes of the two proposed adaptive controlled systems are similar. They are depicted in Fig. 1. Of course, in this figure, the equations of reaction rate estimator, the structure and the adaptive algorithms are in accordance with the two specified cases.

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IV. Simulation Scenarios and Results

The time behaviour of the two adaptive control laws (24) and (25), by comparison to the LCL (11) (the benchmark) was investigated via several simulations. These were accomplished by using the process model described by the relations (1)-(10), for which the yield and kinetic coefficients have the following values [7]:

$\nu^0_{P\max} = 2.613\,h^{-1}$, $\mu^0_{\max} = 0.41\,h^{-1}$, $Y_{PS} = 0.495$, $\gamma = 8.77$, $\alpha = 21.05$, $\beta = 0.824$, $\delta = 0.0366$, $P_{ma} = 217\,g/l$, $P_{ob} = 50\,g/l$, $P_{mb} = 108\,g/l$, $K_S = 0.5\,g/l$, $K_i = 200\,g/l$, $S_i = 80\,g/l$, $P_{me} = 127\,g/l$, $K_{mp} = 0.5\,g/l$, $r = 0.25$, $a = 0.142$, $b = 1.415$.

As it was revealed in [8], [9], this kind of bioprocess exhibits (in open loop) an oscillatory behaviour. Thus, our aim is to design useful control algorithms able to stabilize the bioprocess and to keep the substrate concentration at certain set-point, such that to have a maximum alcohol production.

The following two realistic simulation scenarios were implemented:

Scenario 1. The time behaviour and performance of the adaptive control algorithm (24) is examined, by comparison to LCL (11). The simulation conditions are: C_X, I and Z are not measured; μ_X and ν_P are unknown; C_{Sin} is time varying (see Fig. 2) and unknown; the concentrations C_S and C_P are measurable; the other kinetic and process coefficients are constant and known. Also, $\mu_{\max}$ and $\nu_{P\max}$ are time varying:

$$\mu_{\max}(t) = \mu^0_{\max}(1 + 0.2\cos(\pi t/25)), \quad (26)$$

$$\nu_{P\max}(t) = \nu^0_{P\max}(1 + 0.2\sin(\pi t/20)). \quad (27)$$

Although $\mu_{\max}$ and $\nu_{P\max}$ can be considered as intrinsic properties for some particular microorganism growth, here these have been assumed time varying only in order to consider the kinetic rates as time varying parameters in order to test the robustness of the proposed control strategies.

The system behaviour in this case is represented in Figs. 3-6. The plots from Fig. 3 show the controlled variable C_S, and in Fig. 4 the control input d_S is presented. The set-point C_{Sref} has a desired trajectory such that to reach a low level via several changes as follows: $C_{Sref} = 35\,g/l$, for $t \in [0, 90]$ h, $C_{Sref} = 30\,g/l$, for $t \in (0, 170]$ h, and $C_{Sref} = 25\,g/l$, for $t \in (170, 240]\,h$.

For the first 4 hours the bioprocess runs in open loop; the control input is kept constant at $d_S = 0.08\ h^{-1}$. After that, the loop is closed by using (11) in the LCL case, and (24) in the adaptive case. To be in accordance with the practice, we consider that the measurements of C_S and C_P are contaminated with a zero average additive white noise (2.5% from its nominal values). In Figs. 3 and 4 it can be seen that C_S follows the imposed reference, and the control input d_S is kept in certain limits, i.e. it is limited both superior at a value ($0.08\ h^{-1}$), and inferior to value closely to 0.

The time profile of C_{Sin} estimates delivered by the SMO (14)-(15) is presented in Fig. 5. It is visible that the SMO performs in a proper manner, although the available measurements are influenced by noise, the variation of C_{Sin} is large and its dynamics is fast. The estimates of parameter $\theta_1(t)$ provided by OBE (20)-(21) are given in Fig. 6.

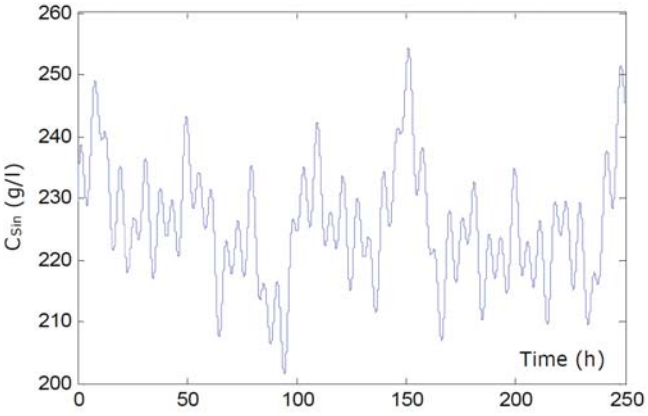


Fig. 2. Time profile of C_{Sin}.

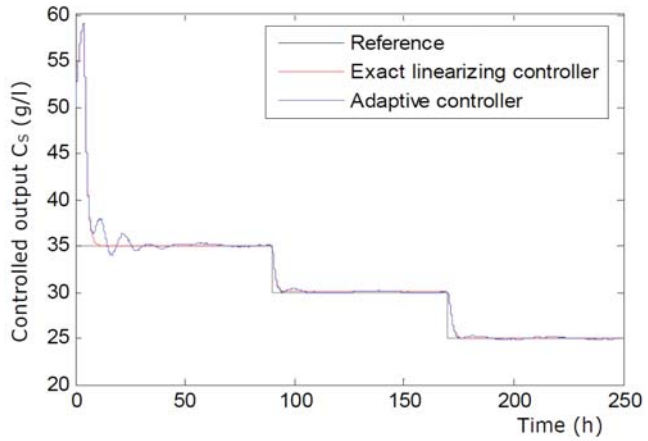


Fig. 3. Evolution of output C_S – first scenario.

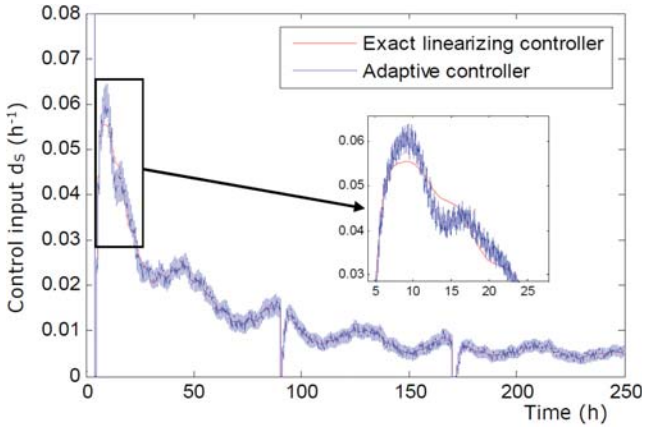


Fig. 4. Time profile of control input d_S – scenario 1.

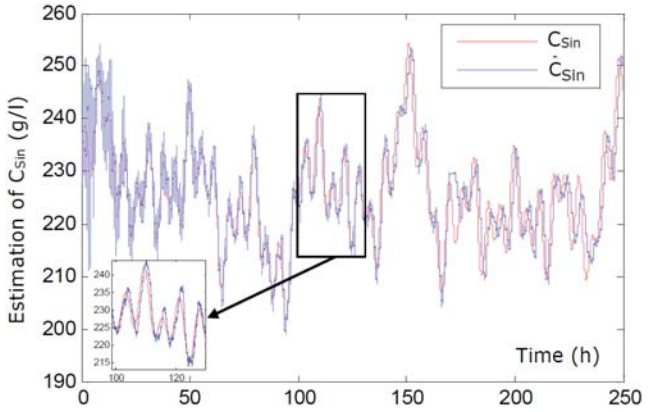


Fig. 5. Estimates of unknown variable C_{Sin} – scenario 1.

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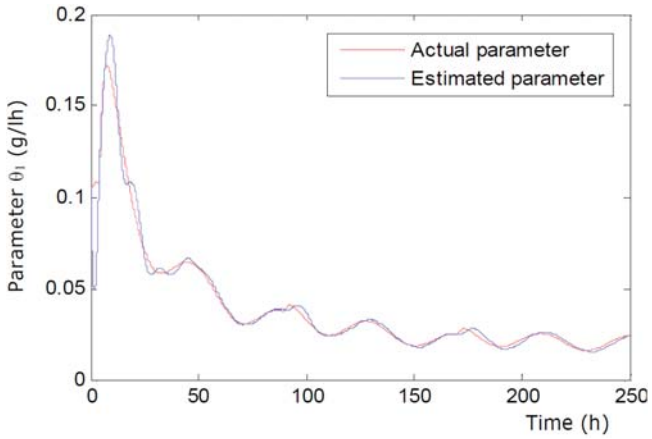


Fig. 6. Estimation of unknown parameter θ_1.

The gain η from control laws (24) and (11) and the tuning parameters of the OBE (20)-(21) are: $\eta = 0.8$, $\lambda_1 = -0.25$, $\omega_1 = 0.00005$. For the SMO (14)-(15), the values of the tuning parameters are: $\lambda_\psi = 0.225$, $\kappa_1 = 0.5$, $\kappa_2 = 5000$, $\gamma_1 = 0.005$, $\gamma_2 = 15$.

From Figs. 3-6 it results that the adaptive controlled process (Scenario 1) behaves well, being very close to the ideal case when the exact LCL (11) is implemented (known bioprocess model). The controller is able to maintain the concentration C_S (the output) very close to the set-point, despite the time variation of C_{Sin} and of several process parameters.

Scenario 2. The closed-loop system obtained by using the adaptive controller (25) will be evaluated, under the same conditions as in the first scenario: C_{Sin} is time varying (as in Fig. 2) and assumed to be unknown; μ_X and ν_P are unknown; C_X, Z and I are not measurable; C_S and C_P are on-line measurable; the other kinetic and process coefficients are known (and constant); the parameters μ_{max} and ν_{Pmax} are time varying as in the first scenario. The behaviour of closed-loop system by comparison to the LCL is presented in Figs. 7-11. In this simulation scenario, the gain η of the controllers (25) and (11), and the tuning parameters of the OBE (20)-(21) are (obtained and chosen as it was abovementioned): $\eta = 0.8$, $\lambda_1 = -0.25$, $\omega_1 = 0.00005$. For the SMO (14)-(15), the tuning parameters were set to: $\lambda_\psi = 0.225$, $\kappa_1 = 0.5$, $\kappa_2 = 5000$, $\gamma_1 = 0.005$, $\gamma_2 = 15$.

Fig. 7 presents the evolution of the process output C_S, and Fig. 8 shows the control input d_S, with similar behaviour as in the first scenario (Figs. 3 and 4). The profile of the estimates of C_{Sin} delivered by the SMO (14)-(15) plotted in Fig. 9 shows that the SMO performs well even if the variation of C_{Sin} is large, its dynamics is fast, and the measurements are affected by noise. Also, the graphics from Fig. 10 that depict the estimates of the unknown parameter $\theta_2(t)$, provided by OBE (20)-(21), show an acceptable behaviour of this estimator.

From the time profiles provided in Figs. 7-8 it can be remarked that the time behaviour of the adaptive controller (25) is good, even better than the behaviour described in the first scenario, although a smaller quantity of a priori information is used.

Fig. 11 presents the time profile of the ethanol concentration (which is quite similar for the two simulation

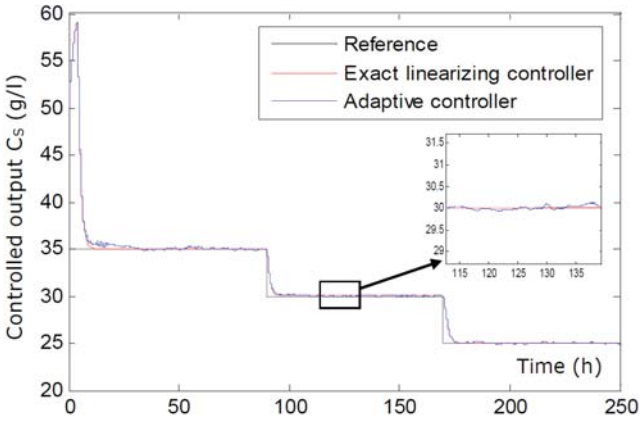


Fig. 7. Evolution of controlled output – scenario 2.

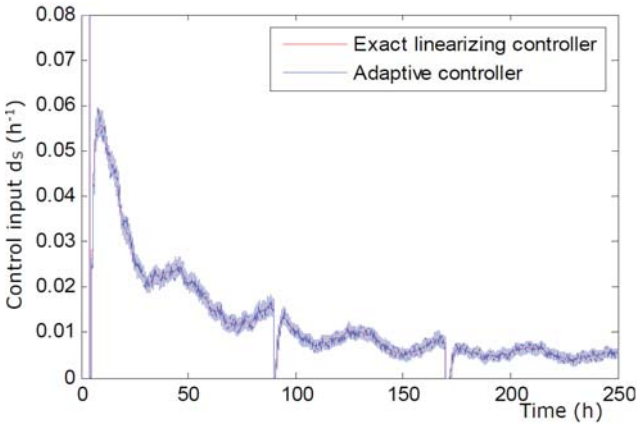


Fig. 8. Time profile of control input – second scenario.

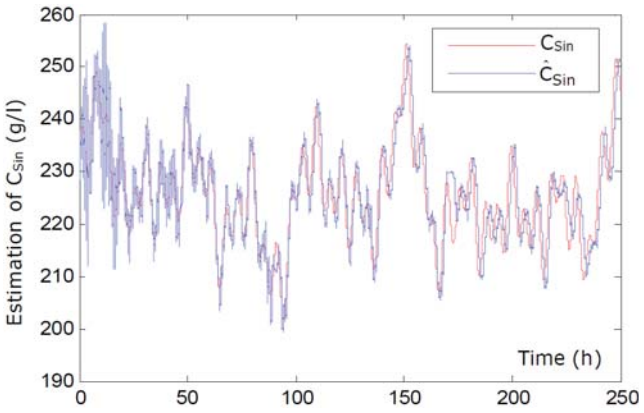


Fig. 9. Estimation of the unknown variable C_{Sin} – scenario 2.

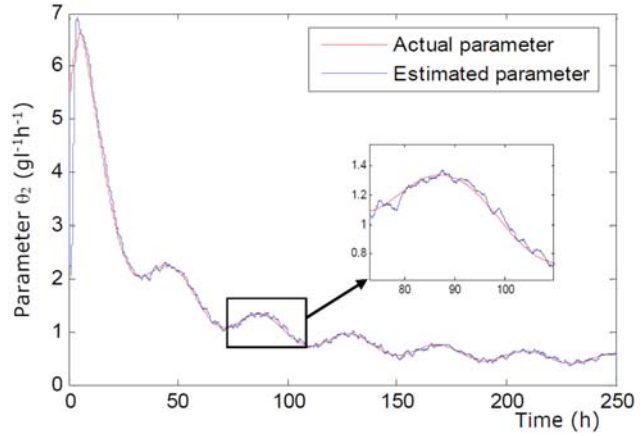


Fig. 10. Estimates of the unknown parameter θ_2.

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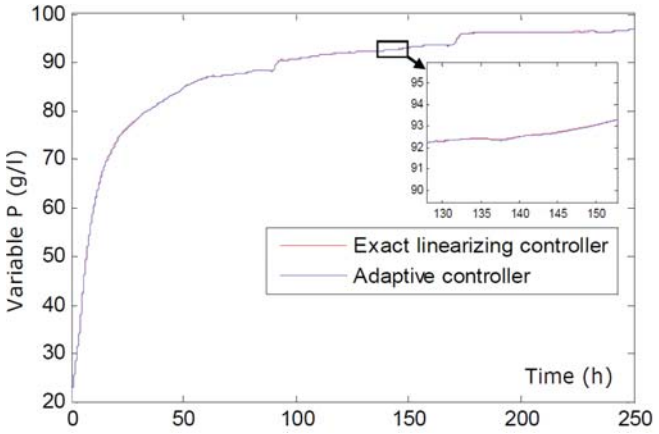


Fig. 11. Evolution of the ethanol concentration.

scenarios). At this point, it is important to remark the important connection between the evolution of residual substrate (Figs. 3 and 7, respectively) and the time profile of product concentration (Fig. 11): if the residual substrate level is kept at a low level, a high concentration of alcohol is obtained.

V. Conclusion

Two innovative adaptive control techniques for a continuous AFP were designed and widely analysed. The design was accomplished under the assumptions that the concentration of influent substrate is unknown (as in the real plants) and the bioprocess kinetics are strongly nonlinear and unknown (partial or total).

Each designed adaptive structure includes a linearizing control law, a sliding mode observer which is used to provide the unknown input concentration, as well as a suitable parameter estimator, utilised for the estimation of the unknown process kinetics.

The simulations were implemented under severe conditions: the concentration of the influent concentration (considered unknown) varies in large limits with fast dynamics, and the process measurements are affected by noise. However, under these conditions, the simulation results showed that the time behaviour of the controlled AFP is accurate, and the control goal was accomplished in both proposed adaptive control schemes.

References

[1] D. Gaida, C. Wolf, and M. Bongards, "Feed control of anaerobic digestion processes for renewable energy production: A review," Renew. & Sust. Energy Reviews, vol. 68, part 2, pp. 869–875, 2017.

[2] O. Bernard, "Hurdles and challenges for modelling and control of microalgae for CO_2 mitigation and biofuel production," J. Process Contr. vol. 21, pp. 1378–1389, 2011.

[3] P.L. Rogers, Y.J. Jeon, K.J. Lee, and H.G. Lawford, "Zymomonas mobilis for fuel ethanol and higher value products," Biofuels, in: Olsson L. (Ed.) Adv. Biochem. Eng. Biotechnol., vol. 108, pp. 263–288, 2007.

[4] E. Petre and D. Selişteanu, Adaptive and robust-adaptive control schemes for an anaerobic bioprocess with biogas production, in: Proc. of the 17th Int. Conf. on System Theory, Control and Computing - ICSTCC 2013, Sinaia, Romania, 2013, pp. 404–409.

[5] E. Petre, D. Selişteanu, and D. Şendrescu, "Adaptive and robust-adaptive control strategies for anaerobic wastewater treatment bioprocesses," Chem. Eng. J., vol. 217, pp. 363–378, 2013.

[6] O. Quintero, A.A. Amicarelli, G. Scaglia, and F. Sciascio, "Control based on numerical methods and recursive bayesian estimation in a continuous alcoholic fermentation bioprocess," Bioresources, vol. 4, no. 4, pp. 1372–1395, 2009.

[7] O. Quintero, A.A. Amicarelli, F. Sciascio, and G. Scaglia, "State estimation in alcoholic fermentation bioprocess of Zymomonas Mobilis using recursive Bayesian filtering: A simulation approach," Bioresources, vol. 3, no. 2, pp. 316–334, 2008.

[8] A.J. Daugulis, P.J. McLellan, and J. Li, "Experimental investigation and modeling of oscilllatory behavior in the continuous culture of Zymomonas mobilis", Biotechnol. Bioeng., vol. 56, pp. 99-105, 1997.

[9] I. Oliveira Pinheiro, A.C. Rapôso Camêlo, A.M. Souto-Maior, and C.E. Lopes, "Oscillatory behavior of zymomonas mobilis in continuous stirred tank bioreactor," 2nd Mercosur Congress on Chemical Eng., 4th Mercosur Congress on Process Systems Eng., Aug. 14-18, 2005, ENPROMER, Rio de Janeiro, Brasil, pp. 1-10.

[10] G. Bastin and D. Dochain, On-line Estimation and Adaptive Control of Bioreactors. Amsterdam: Elsevier, 1990.

[11] P.A. López-Pérez, F.A. Cuevas-Ortiz, R.V. Gómez-Acata, and R. Aguilar-López, "Improving bioethanol production via nonlinear controller with noisy measurements," Chem. Eng. Commun., vol. 202, no. 11, pp. 1438–1445, 2015.

[12] A. Ajbar and E. Ali, "Study of advanced control of ethanol production through continuous fermentation," Journal of King Saud University - Engineering Sciences, vol. 29, Issue 1, pp. 1–11, 2017.

[13] W. Herrera, E. Rivera, L.A. Alvarez, L. Plazas Tovar, S. Tamayo Rojas, C. Yamakawa, A. Bonomi, and R. Maciel Filho, "Modelling and control of a continuous ethanol fermentation using a mixture of enzymatic hydrolysate and molasses from sugarcane," Chem. Eng. Transactions, vol. 50, pp. 169–174, 2016.

[14] A. Azadeh, H.V Arani, and H. Dashti, "A stochastic programming approach towards optimization of biofuel supply chain," Energy, vol. 76, pp. 513–525, 2014.

[15] E. Petre and D. Selişteanu, "A robust-adaptive control strategy for a continuous alcoholic fermentation process," in: Proc. of the 18th Int. Conf. on System Theory, Control & Computing, Sinaia, Romania, 2014, pp. 424–429.

[16] E. Petre, D. Selişteanu, D. Şendrescu, M. Barbu, and S. Caraman, "An adaptive control structure for an anaerobic digestion process with unknown inputs," in: Proc. of the 18th Int. Carpathian Control Conf. (ICCC'2017), Sinaia, Romania, 2017, pp. 58–63.

[17] M. Sbarciog, G. Giovannini, R. Chamy, and A. Vande Wouwer, "Control and estimation of anaerobic digestion processes using hydrogen and volatile fatty acids measurements," Water Sci. Technol., vol. 78, no. 10, pp. 2027–2035, 2018.

[18] E. Petre, D. Selişteanu, and M. Roman, "Nonlinear robust adaptive control strategies for a lactic fermentation process," J. Chem. Technol. Biotechnol., vol. 93, pp. 518–526, 2018.

[19] E. Petre, M. Roman, and D. Selişteanu, "Nonlinear estimation and control schemes for a complex anaerobic digestion of microalgae with unknown kinetics and inputs," Bioresource Technol., vol. 287, 121429, 2019.

[20] J.A. Moreno, "On discontinuous observers for second order systems: Properties, analysis and design," in: B. Bandyopadhyay, S. Janardhanan, S. Spurgeon (eds) Advances in Sliding Mode Control. Lecture Notes in Control and Information Sciences, vol. 440, 2013, Springer, Berlin, pp. 243–265.

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